

# Representativeness: A Possibility Theorem

Mateus Hiro Nagata

HEC Paris

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## Introduction

## Case-Based Decision Theory

## Representative Selection as a Learning Problem

## Asymptotic Representational Equivalence

## Conclusion

$\succsim$  on policies, vote on representatives

# The Representative Dream Problem

## Dream Problem

Candidates + data on past problems & actions  $\rightarrow$  find one close to society's preferences, in- & out-of-sample?

- ▶ Future problems: unknowable at election time
- ▶ Savage (1954): probabilities over all states (grand-world)  $\rightarrow$  cognitively implausible
- ▶ Case-Based Decision Theory

## This paper

Reframe representative selection as a **statistical learning problem** and prove it is efficiently solvable.

# This Paper

**Main question:** 1) If I see enough actions of a candidate, can I predict its next action? 2) If there are enough candidates, can I find one that acts very close to me? **Yes:**

1. **Finite-sample bound:** excess risk  
$$\leq 2\sqrt{2 \ln n / m} + 5\sqrt{2 \ln(8/\delta) / m}$$
2. **Sample complexity:**  $m = O\left(\frac{\ln n}{\epsilon^2}\right)$  - only logarithmic in the number of candidates
3. **Asymptotic:** under parametric similarity + dense coverage,  
 $L_D(i^*) \xrightarrow{P} 0$

**Contribution:** A *possibility theorem* for representative democracy.

## Related Literature

- ▶ **Political economy:** Besley & Coate (1997), Osborne & Slivinski (1996) citizen-candidate games; do not address sample complexity
- ▶ **Social choice:** Arrow (1950), Gibbard (1973), Satterthwaite (1975) — impossibility of preference *aggregation*; we study preference *estimation*
- ▶ **Preference learning:** Fürnkranz & Hüllermeier (2010), Ailon & Mohri (2010) — ranking from pairwise comparisons; no representative selection with distributional guarantees
- ▶ **CBDT:** Gilboa & Schmeidler (1995), Billot et al. (2008) — individual reasoning; we apply it to collective representation
- ▶ **Learning theory:** Valiant (1984), Vapnik (2013), Bartlett & Mendelson (2002) — standard tools applied to a novel political problem

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## Memory & Similarity

**Shared memory:**  $\mathcal{M} = \{(q, a, r)\}$ : problem, action, outcome.

**History:**  $H = \{q \in P \mid \exists a, r: (q, a, r) \in \mathcal{M}\}$

New problem  $p \notin H \rightarrow$  decide via similarity to past cases.



# Case-Based Decision Theory

## Theorem (Representation)

*A1-A5, then  $\exists s : P \times P \rightarrow [0, 1]$  and  $u : \mathbb{R} \rightarrow \mathbb{R}$ , both unique up to scaling, s.t.*

$$x \succsim_{p,H} y \iff \sum_{q \in H} s(p, q) u(x(q)) \geq \sum_{q \in H} s(p, q) u(y(q))$$

**Decision rule:**

$$V_i(a \mid p, \mathcal{M}) = \sum_{\substack{(q, a', r) \in \mathcal{M} \\ a' = a}} s_i(p, q) \cdot u(r)$$

$$a^i(p, \mathcal{M}) = \arg \max_{a \in A} V_i(a \mid p, \mathcal{M})$$

# Assumptions

## Assumption 1 (Single Shared Memory)

All agents observe the same  $\mathcal{M}$ . No private information about past outcomes.

## Assumption 2 (Common Utility)

All agents agree on outcome evaluation  $u(\cdot)$ ; they differ only in similarity assessments  $s_i$ .

Political disagreement = *different beliefs about relevance* (similarity), not different values.

⇒ Selection is: **identify the candidate  $i$  whose  $s_i$  is closest to  $s_{\text{soc}}$ .**

## Running Example: Fiscal Policy

$A = \{\text{stimulus, austerity}\}$ . New problem  $p_{2026}$ : 115% debt/GDP + financial stress.

| Case              | Action    | $r$ | $s(p_{2026}, q)$ |
|-------------------|-----------|-----|------------------|
| 2008 Crisis       | Stimulus  | +5  | 0.7              |
| 2010 Greece       | Austerity | -3  | 0.4              |
| 1930 Depression   | Austerity | -8  | 0.5              |
| 1970s Stagflation | Stimulus  | -2  | 0.2              |

$$V(\text{stim}) = 0.7(5) + 0.2(-2) = 3.1$$

$$V(\text{auster}) = 0.4(-3) + 0.5(-8) = -5.2 \quad (V(\text{stim}) > V(\text{auster}))$$

Society  $\rightarrow$  **stimulus**. A ( $s_A(p_{2026}, q_{2008}) = 0.9$ ) agrees; B (debt-focused) doesn't.

*Which generalises?*

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## The Selection Rule

Voters observe  $m$  problems  $S = \{p_1, \dots, p_m\} \stackrel{\text{i.i.d.}}{\sim} D$  and candidates' choices  $\{a_j^i\}$ .

**Empirical disagreement** (observable):

$$L_S(i) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j^i \neq a_j^{\text{soc}}]$$

**Population risk** (target, unobservable):

$$L_D(i) = \Pr_{p \sim D}[a^i(p, \mathcal{M}) \neq a^{\text{soc}}(p, \mathcal{M})]$$

**ERM selection rule:**

$$i^* = \arg \min_{i \in \{1, \dots, n\}} L_S(i)$$

**Question:** Does low  $L_S(i^*)$  imply low  $L_D(i^*)$ ?

# Theorem 1: Finite-Sample Guarantee

## Theorem (Uniform Convergence)

With probability  $\geq 1 - \delta$ :

$$L_D(i^*) - \min_i L_D(i) \leq 2\sqrt{\frac{2 \ln n}{m}} + 5\sqrt{\frac{2 \ln(8/\delta)}{m}}$$

- ▶ Best candidate *in the list*  $\Rightarrow$  generalizes out-of-sample
- ▶  $n$  enters only as  $\ln n$ :  $10 \rightarrow 100$  candidates  $\approx 2\times$  data, not  $10\times$

## Interpretation: Theorem 1 & Fiscal Policy

- ▶  $n = 100$  candidates, each with own  $(s_i, w_i)$  over debt/growth/etc.
- ▶ Observe  $m$  past recessions: each candidate's stance + society's actual choice
- ▶ ERM  $\rightarrow$  candidate who most often “called it right”
- ▶ Thm 1: that candidate's *future* call-it-right rate  $\approx$  best possible — without ever observing the best directly
- ▶ Distribution-free: holds whatever the  $s_i$ 's are, however ideologically diverse
- ▶ Voters needn't know the “right” similarity function ex ante — it's revealed by track record

## Corollary: Sample Complexity

### Corollary

For any  $\varepsilon, \delta \in (0, 1)$ , if

$$m \geq \frac{98}{\varepsilon^2} \left[ \ln n + \ln \frac{8}{\delta} \right]$$

then with probability  $\geq 1 - \delta$ :  $L_D(i^*) \leq \min_i L_D(i) + \varepsilon$ .

**Concrete numbers** ( $\varepsilon = 0.10$ ,  $\delta = 0.05$ ,  $n = 100$ ):  $m \approx 95,000$ .

- ▶ High, but distribution-free (worst-case). Section 4 shows constants drop under parametric structure.
- ▶ **Economic implication:** Candidate diversity is informationally cheap. Restricting ballot access cannot be justified on informational grounds.



## Interpretation: Diversity & Ballot Access

- ▶ 100  $\rightarrow$  1000 candidates: only  $\ln(10) \approx 2.3\times$  more data
- ▶ Against “too many choices confuse voters”: complexity grows log, not linearly
- ▶  $m \approx 95,000$  = many issue-level observations (roll-call votes, policy debates over years) — not 95k elections
- ▶ Caveat: worst case,  $n$  *arbitrary* candidates, no structure — §4 imposes structure  $\rightarrow$  far smaller effective requirement

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## Can representation be *perfect*, asymptotically?

**Exponential similarity** (Billot et al., 2008): shift inv. + ray mono. + symmetry + ray-shift inv. + self-relevance  $\Rightarrow$

$$s(p, q) = e^{-\nu(p-q)}, \quad \nu : \mathbb{R}^d \rightarrow \mathbb{R}_+ \text{ a norm}$$

**Linear utility:**  $u(r, \mathbf{w}) = \langle \mathbf{w}, r \rangle$ ,  $\mathbf{w} \in \Delta^k$  — agree on dimensions (growth, debt, employment, equality), differ in weights.

### A3 — Dense Candidate Coverage

$\{\theta^1, \dots, \theta^n\} = \{(\nu^i, \mathbf{w}^i)\}$  forms an  $\eta_n$ -net of  $\Theta = \{\nu\} \times \Delta^k$ ,  
 $\eta_n \rightarrow 0$ .

*Fiscal policy:* candidates differ in how they weigh problem-distance ( $\nu$ ) and outcome priorities ( $\mathbf{w}$ ).

## Theorem 2: Asymptotic Zero Risk

### Lemma (Lipschitz Continuity)

*Under exponential similarity, linear utility, bounded problems and outcomes, and a margin condition, there exists  $C > 0$  such that*

$$|L_D(i) - L_D(j)| \leq C \|\theta^i - \theta^j\|$$

### Theorem (Asymptotic Representational Equivalence)

*Suppose Assumptions 1–3 hold and  $m(n)/\ln n \rightarrow \infty$ . Then*

$$L_D(i^*) \xrightarrow{p} 0 \quad \text{as } n \rightarrow \infty.$$

**Proof:**  $L_D(i^*) \leq \underbrace{C\eta_n}_{\rightarrow 0 \text{ (coverage)}} + \underbrace{O\left(\sqrt{\ln n/m}\right)}_{\rightarrow 0 \text{ (data)}}.$

## Interpretation: Theorem 2 & Fiscal Policy

- ▶ Pool grows (more parties, lower entry costs) + history accumulates
- ▶  $\Rightarrow \exists i: (\nu^i, \mathbf{w}^i) \approx (\nu^{\text{soc}}, \mathbf{w}^{\text{soc}})$ , arbitrarily close
- ▶ The candidate's stimulus/austerity calls match society's on (almost) every future recession
- ▶ Representation is as good as direct democracy (with diversity and memory)

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## Failure Modes

1. **Sparse coverage** ( $\eta_n \not\rightarrow 0$ ): filing fees, Duverger's Law  $\Rightarrow$  only “centrist” stimulus/austerity on the ballot
2. **Insufficient data** ( $m/\ln n \not\rightarrow \infty$ ): new democracies, destroyed records
3. **Heterogeneous reasoning**: disagree on *how* to reason
4. **Distribution shift**:  $D_t$  moves fast  $\Rightarrow$  history loses predictive power

## Conclusion

- ▶ Representative selection = **estimation**, not aggregation ( $\neq$  Arrow)
- ▶ Possibility theorem:  $O(\ln n/\varepsilon^2)$  observations suffice
- ▶ Asymptotically: dense coverage + growing data  $\Rightarrow$  disagreement prob.  $\rightarrow 0$
- ▶ Institutional implications: low ballot barriers, institutional memory, stationarity — *necessary*, not merely nice-to-have
- ▶ Limitations: static, i.i.d. problems, common utility (base model)



Salut!

Questions?

## What I Would Like to Add

- ▶ Dynamic model: finite terms, re-estimation at each election
- ▶ Non-i.i.d.:  $\beta$ -mixing sequences, distribution shift ( $K$ -bounded density ratios)
- ▶ Endogenous candidate entry: strategic similarity mimicry
- ▶ Empirical application: roll-call data as proxy for  $s$
- ▶ RLHF connection (Dai & Fleisig, 2024): voters = reward signal, candidates = policies

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## Appendix

## Appendix: Proof of Theorem 1 (II)

**Step 2 (McDiarmid).** Bounded differences  $\Rightarrow$  w.p.  $\geq 1 - \delta/2$ :

$$\sup_i |L_D(h^i) - L_S(h^i)| \leq \mathbb{E}_S[\hat{\mathfrak{R}}_S(\mathcal{F})] + \sqrt{\frac{\ln(4/\delta)}{2m}}$$

**Step 3 (ERM bound).** Standard argument  $\Rightarrow$  w.p.  $\geq 1 - \delta$ :

$$L_D(i^*) - \min_i L_D(i) \leq 2\mathbb{E}_S[\hat{\mathfrak{R}}_S(\mathcal{F})] + 5\sqrt{\frac{2\ln(8/\delta)}{m}}$$

**Step 4.** Substitute Step 1:

$$L_D(i^*) - \min_i L_D(i) \leq 2\sqrt{\frac{2\ln n}{m}} + 5\sqrt{\frac{2\ln(8/\delta)}{m}} \quad \blacksquare$$

## Appendix: Proof of Sample Complexity Corollary

From Theorem 1, excess risk  $\leq \sqrt{\frac{2}{m}} \left( 2\sqrt{\ln n} + 5\sqrt{\ln(8/\delta)} \right)$ .

**Cauchy–Schwarz:**

$$\left( 2\sqrt{\ln n} + 5\sqrt{\ln \frac{8}{\delta}} \right)^2 \leq (2^2 + 5^2) (\ln n + \ln \frac{8}{\delta}) = 29 (\ln n + \ln \frac{8}{\delta})$$

$$\Rightarrow \text{excess risk} \leq \sqrt{\frac{58}{m} \left( \ln n + \ln \frac{8}{\delta} \right)}$$

Set  $\leq \varepsilon$ , solve for  $m$ :

$$m \geq \frac{58}{\varepsilon^2} \left\lceil \ln n + \ln \frac{8}{\delta} \right\rceil$$

$$\Rightarrow \text{a fortiori, } m \geq \frac{98}{\varepsilon^2} \left\lceil \ln n + \ln \frac{8}{\delta} \right\rceil \text{ suffices. } \blacksquare$$

(Cauchy–Schwarz gives a tighter constant, 58, than separately bounding each term.)

## Appendix: CBDT Axioms (A1–A5) — Formal

Fix  $p \in \mathcal{P}$ , history  $H$ . Profiles  $x, y, z, w \in \mathbb{R}^H$ ;  $v_q$  is the unit vector at  $q \in H$ .  
 “Compatible” = jointly admissible under memory.

### Axiom (Comparability)

For all compatible  $x, y \in \mathbb{R}^H$ :  $x \succeq_{p,H} y$  or  $y \succeq_{p,H} x$ .

### Axiom (Monotonicity)

If  $x, y$  compatible,  $x \cdot y = 0$ , and  $x(q) \geq y(q)$  for all  $q \in H$ , then  $x \succeq_{p,H} y$ .

### Axiom (Continuity)

$\{y : y \succeq_{p,H} x\}$  and  $\{y : x \succeq_{p,H} y\}$  are closed in  $\mathbb{R}^H$ .

### Axiom (Separability)

For all  $x, y, z, w \in \mathbb{R}^H$  with  $(x+z), (y+w)$  compatible: if  $x \succeq_{p,H} y$  and  $z \succeq_{p,H} w$ , then  $(x+z) \succeq_{p,H} (y+w)$ ; strictly if either antecedent is strict.

### Axiom (Similarity Invariance)

For memories  $M_1, M_2$  with  $q_1, q_2 \in H_i$ ,  $p \notin H_i$ , and unit vectors  $v_j^i$ : if  $x \sim_{p,H_1} y$ ,  $z \sim_{p,H_2} w$ , and  $x + \alpha v_1^1 \sim_{p,H_1} y + \beta v_2^1$ , then  $z + \alpha v_1^2 \sim_{p,H_2} w + \beta v_2^2$  (whenever  $\alpha, \beta \geq 0$ ).

## Appendix: CDBT Axioms — Interpretation

**Representation (Gilboa & Schmeidler, 1995).**  $A1-A5 \Rightarrow \exists s, u$ :

$$x \succsim_{p,H} y \iff \sum_q s(p, q) u(x(q)) \geq \sum_q s(p, q) u(y(q))$$

$s$  unique up to scaling,  $u$  up to positive affine transform.

### A1–A4 (regularity):

- ▶ **Comparability:** voters can always compare candidates' records
- ▶ **Monotonicity:** better outcomes  $\rightarrow$  weakly preferred (agreement on  $u$ , relaxed in §4)
- ▶ **Continuity:** rules out single-issue / lexicographic voting
- ▶ **Separability:** fiscal record judged independently of foreign policy

**A5 — the substantive one:** relative weight of  $q_1$  vs.  $q_2$  is intrinsic to  $(p, q_1, q_2)$ , not memory-dependent. Each  $s_i$  is a **fixed, portable trait** — precisely what lets observed historical agreement generalise to future agreement (Thm 1).



## Appendix: Exponential Similarity Axioms — Formal

Similarity  $s : \mathcal{P} \times \mathcal{P} \rightarrow (0, 1]$ ,  $\mathcal{P} \subseteq \mathbb{R}^d$ .

### Axiom (Shift Invariance)

*For all  $p, q$  and  $v \in \mathbb{R}^d$  with  $p + v, q + v \in \mathcal{P}$ :  $s(p, q) = s(p + v, q + v)$ .*

### Axiom (Ray Monotonicity)

*For all  $p$  and  $v \in \mathbb{R}^d \setminus \{0\}$ :  $t \mapsto s(p, p + tv)$  is decreasing in  $t > 0$ .*

### Axiom (Symmetry)

*For all  $p, q$ :  $s(p, q) = s(q, p)$ .*

### Axiom (Ray Shift Invariance)

*For all  $p, q$  and  $\lambda > 0$ :  $s(p, \lambda p) = s(q, \lambda q)$  whenever both are defined.*

### Axiom (Self-Relevance)

*For all  $p$ :  $s(p, p) = 1$ .*

### Theorem (Billot et al., 2008)

## Appendix: Exponential Similarity — Interpretation

- ▶ **Shift inv.:** only the *gap* matters — 3% vs 5% debt  $\equiv$  103% vs 105%
- ▶ **Ray mono.:** 110% debt is closer to 100% than to 120%
- ▶ **Symmetry:** 2008 informs 2026 as much as 2026 would inform 2008
- ▶ **Ray-shift inv.:** no special / privileged region of  $P$
- ▶ **Self-relevance:** normalises the scale ( $s = 1 \Leftrightarrow$  identical)

$\Rightarrow$  political-economy analogue of a Gaussian/RBF kernel; matches Shepard's (1987) law of similarity decay in psychology — people reason by graded analogy, not sharp categories.

**If challenged:** CBDT alone (any  $s$ ) suffices for Thm 1 / sample complexity. Exponential similarity is needed only for Thm 2's Lipschitz argument — any continuously-parametrised  $s$  gives the same conclusion.